

ECE 186: Homework 1

- A = answering question 1 & winning $E[A] = 0.8 \cdot 100 = \$80$
 B = answering question 2 & winning $E[B] = 0.5 \cdot 200 = \$100$
 Choose question 2 to maximize expected reward.

- $$F_X(x) = P(X \leq x) = \frac{2x^2}{2r^2} = \left(\frac{x^2}{r^2}\right)$$

$$f_X(x) = \frac{d}{dx} F_X(x) = \left(\frac{2x}{r^2}\right)$$

$$E[X] = \int_0^r \frac{2x^2}{r^2} dx = \frac{2x^3}{3r^2} \Big|_0^r = \frac{2r^3}{3r^2} = \left(\frac{2}{3}r\right)$$

$$E[X^2] = \int_0^r \frac{2x^3}{r^2} dx = \frac{2x^4}{4r^2} \Big|_0^r = \frac{r^4}{2r^2} = \frac{r^2}{2}$$

$$\text{Var}(X) = \frac{r^2}{2} - \left(\frac{2}{3}r\right)^2 \Rightarrow \frac{r^2}{2} - \frac{4}{9}r^2 = \left(\frac{r^2}{18}\right)$$

- $$P(Z > z) = P(\min(X, Y) > z) = P(X > z) P(Y > z)$$

$$= e^{-z} \cdot e^{-z} = \left(e^{-2z}\right)$$

$$F_Z(z) = P(Z \leq z) = 1 - e^{-2z} \quad \text{for } z \geq 0$$

$$f_Z(z) = 2e^{-2z} \quad \text{for } z \geq 0$$

Z is an exponential random variable.

- a) $\frac{S_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} E[X] \quad E[X] = \int_0^1 2x^2 dx = \frac{2x^3}{3} \Big|_0^1 = \frac{2}{3}$

$$\left[\lim_{n \rightarrow \infty} \frac{S_n}{n} = \frac{1}{E[X]} = 3/2 \text{ components per day} \right]$$

- b) $E[X^2] = \int_0^1 2x^3 dx = \frac{2x^4}{4} \Big|_0^1 = \frac{1}{2}$

$$\text{Var}(X) = \frac{1}{2} - \frac{4}{9} = \frac{9}{18} - \frac{8}{18} = \frac{1}{18}$$

$$P(S_n \geq 365) = 0.95$$

$$\frac{365 - \frac{2}{3}n}{\sqrt{\frac{1}{18}}} = -2 \quad \quad 365 - \frac{2}{3}n = -2 \cdot 0.95 \sqrt{\frac{n}{18}}$$

$$n \approx 561.279 \rightarrow 562 \text{ components}$$